

Supplementary Material

Proof of NP-hardness of the Joint Task Offloading and UAV Trajectory Optimization Problem

This supplementary document provides the full proof of Theorem 1 (NP-hardness) referenced in the main paper. All notation follows the main text.

Problem Statement

We consider the joint task offloading and UAV trajectory optimization problem in a Space–Air–Ground Integrated Network (SAGIN) with mobile user equipments (MUEs), UAVs, LEO satellites, and base stations (BSs). Each MUE $e_i \in E$ has a task $Q_i = (A_i, C_i)$ with data size A_i and required CPU cycles C_i . The offloading decision of MUE e_i is a triplet $o_i = (a_i, b_i, c_i)$, where $a_i \in \{0, 1, 2, 3\}$ selects among local computing, UAV, LEO, or BS, b_i is the server index, and c_i is the wireless channel.

Let $\mathcal{V} = U \cup S \cup B$ denote the set of all edge computing nodes. Define $\mathcal{I}_v^+ = \{i \mid a_i^+ \neq 0, v_i^+ = v\}$ as the set of MUEs offloading to node v . The problem is formulated as

$$\min_{(\mathbf{o}^+, \mathbf{q}^+) \in \Omega(t)} \sum_{i=1}^N K_i(o_i^+, \mathbf{o}_{-i}^+, \mathbf{q}^+), \quad (F_1)$$

subject to LEO communication-window constraints, edge-node computation capacity $\sum_{i \in \mathcal{I}_v^+} f_{i,v}^+ \leq F_v^{\max}$ for $v \in \mathcal{V}$, UAV movement $\|q_j^+ - q_j(t)\|_2 \leq D_{\max}$, individual rationality $K_i \leq K_{e_i}^{\text{local}}$, coverage constraints for UAV and BS, and feasibility of the decision variables.

Theorem 1. *The optimization problem F_1 is NP-hard.*

Proof. We prove NP-hardness by showing that the Generalized Assignment Problem (GAP), a classical NP-hard problem, is a restricted instance of F_1 . Consider a special case of F_1 where UAV positions are fixed, local computing is disabled, channel selection and co-channel interference are ignored, and each MUE must offload to exactly one feasible edge server. Candidate servers include UAVs, LEO satellites, and BSs, each with finite computation capacity.

Given any GAP instance, each job maps to one MUE task and each machine maps to one edge server in $\mathcal{V} = U \cup S \cup B$. Let $x_{ij} \in \{0, 1\}$ denote whether task i is assigned to server j . Assigning task i to server j incurs offloading cost K_{ij} (GAP assignment cost) and consumes r_{ij} units of computation resource. Since UAV positions are fixed and wireless interference is ignored, K_{ij} is constant. Each server j has capacity R_j . Assignments violating LEO visibility, UAV coverage, or BS coverage constraints are infeasible.

Under this restricted setting, the problem becomes

$$\begin{aligned}
& \min_{\{x_{ij}\}} \quad \sum_{i=1}^N \sum_{j \in \mathcal{V}} x_{ij} K_{ij} \\
& \text{s.t.} \quad \sum_{j \in \mathcal{V}} x_{ij} = 1, \quad \forall i, \\
& \quad \sum_{i=1}^N x_{ij} r_{ij} \leq R_j, \quad \forall j \in \mathcal{V}, \\
& \quad x_{ij} = 0, \quad \text{if assignment } (i, j) \text{ is infeasible,} \\
& \quad x_{ij} \in \{0, 1\}, \quad \forall i, j.
\end{aligned} \tag{1}$$

This is a generalized assignment problem with forbidden assignments. Since GAP is NP-hard, this restricted instance of F_1 is NP-hard. The original F_1 further includes local-computing decisions, channel selection, co-channel interference, LEO visibility, UAV/BS coverage, and UAV trajectory optimization, so F_1 is at least as general and NP-hard. $\square$